

EXPERIMENT 19. Perform Edge Detection using Sobel Matrix along XY Axis

Aim

To detect the edges of an image using the **Sobel operator along both X and Y axes**.

Procedure

1. Read the input image.
2. Convert the image into grayscale.
3. Apply the Sobel operator along the **X-axis**.
4. Apply the Sobel operator along the **Y-axis**.
5. Combine the X and Y gradients to obtain the **XY edge image**.
6. Display the original image and detected edges.

Code

```
import cv2
import numpy as np

# Read image
img = cv2.imread("image.jpg")

# Convert to grayscale
gray = cv2.cvtColor(img, cv2.COLOR_BGR2GRAY)

# Sobel X
sobel_x = cv2.Sobel(gray, cv2.CV_64F, 1, 0, ksize=3)

# Sobel Y
sobel_y = cv2.Sobel(gray, cv2.CV_64F, 0, 1, ksize=3)

# Combine X and Y
sobel_xy = cv2.magnitude(sobel_x, sobel_y)
```

Convert to 8-bit

```
sobel_xy = cv2.convertScaleAbs(sobel_xy)
```

Display

```
cv2.imshow("Original Image", img)
```

```
cv2.imshow("Sobel XY Edge Detection", sobel_xy)
```

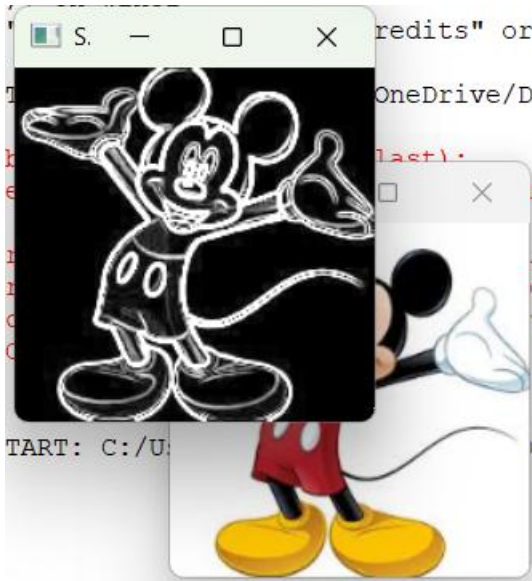
```
cv2.waitKey(0)
```

```
cv2.destroyAllWindows()
```

Input



Output



Result

Thus, the edges of the input image were successfully detected using the **Sobel Matrix along the X and Y axes (XY direction)**.